

Steel Stud Walls: Breaking the Thermal Bridge

by Jan Kosny

Steel stud walls are notoriously inefficient due to thermal bridging, but new construction and insulating methods are solving that problem.

Steel framing has many advantages over wood framing (see “Steel Framing: How Green?” p. 6). But it has one significant disadvantage: Steel studs conduct heat extremely well. This effect is known as thermal bridging, and it can sharply reduce a wall’s effective R-value. The simplest and most common way to overcome this problem is to block the path of heat flow with rigid foam insulation (see Table 1).

Adding rigid foam insulation not only increases the whole wall R-

value, but it also reduces the temperature difference between the center of the cavity and the stud area. This cuts down on the possibility that black stains will form on interior walls from dirt that is attracted to cold spots on the wall surfaces. However, rigid foam insulation is an expensive solution to the problem of thermal bridging in steel stud construction.

In an effort to find more cost-effective solutions, researchers here at Oak Ridge National Laboratory (ORNL) have explored many

options, using both hot-box testing and computer simulations. These options have included diminishing the contact area between the studs and the sheathing, reducing the steel stud web area, replacing the steel web with a less conductive material, and placing foam insulation in locations where the thermal shorts are most critical. While examining these options, I and my colleagues at ORNL’s Building Technology Center (BTC) also strove to develop energy-efficient technologies that would enable

Steel Stud Wall with Vertical Ridges on Stud Flange Areas

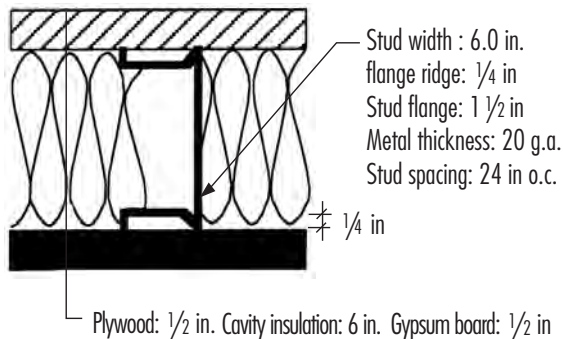


Figure 1. Vertical ridges on a steel stud reduce the contact area between the stud and the sheathing material and improve the whole wall R-value.

steel stud walls to beat the performance of traditional 2 x 6 wood stud walls. We achieved this goal and are now looking for industrial partners who will help to commercialize our novel steel stud technologies.

Reducing Contact

We examined four ways of reducing the contact area between the studs and the sheathing in steel stud walls. These were:

- placing ridges in the stud flange area;
- placing dimples in the stud flange area;
- placing wood or metal spacers in the stud flange area; and
- putting thin foam tape on the face of the stud flanges.

When ridges are created in the surface of the flange, the sheathing material in the wall is not supported by the whole flange, but only by the surface of the ridges. We tested and simulated several walls with vertical distance ridges (see Fig-

ure 1). We found that these ridges reduced the contact area between the studs and the sheathing material by about 95% and improved the whole wall R-value by 9%–12%. These results suggest that the thermal bridge effect was so strong that even significant reductions in steel flange contact area could not proportionally increase the wall's R-value.

We examined two other walls

to investigate the thermal effects of using studs with 1-inch extruded dimples on the flange surfaces. A traditionally constructed 3 1/2-inch steel stud wall was simulated for purposes of comparison. All the walls had 3 1/2-inch studs at 16 inches on center (OC). The wall cavity was insulated with R-11 batts. The extruded dimples reduced the contact area between studs and the sheathing material by 89% and improved the R-value by 9%. We found that the thermal improvement from this technique

was insignificant compared to the scale of design changes.

Distance spacers can also be used to reduce the thermal bridge effect in steel stud walls. To determine whether these spacers would be more effective if they were made of less conductive materials than metal,

Steel Stud with Extruded Dimples

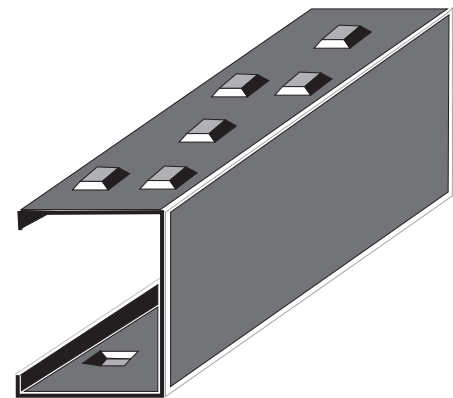


Figure 2. Extruded dimples on a steel stud also reduce contact between the stud and the sheathing and improve the wall R-value.

we installed either horizontal steel or wooden furring strips. These separated the steel stud from the exterior sheathing, creating an air cavity. In another series of hot-box tests and computer simulations, we examined the thermal effects of applying

these spacers to four walls. For all the walls with spacers, the increase in wall R-value was close to the R-value of the additional air space. Results were similar for wood and steel spacers.

Reducing the Steel Stud Web

We also examined the effects of reducing the stud web area (the area that forms a bridge between two steel flanges), or replacing the steel web with less conductive materials. A very intensive heat transfer

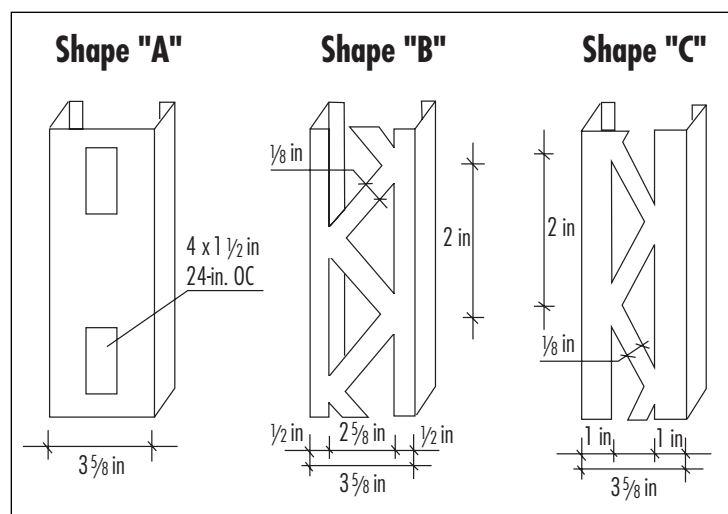


Figure 3. Openings on the stud web area reduce the heat conducting area of the stud web, reducing the thermal bridge effect.

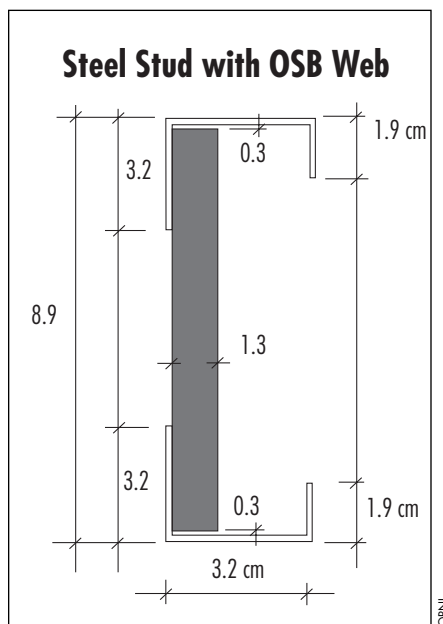


Figure 4. Replacing steel in the stud web with OSB increases the stud's thermal resistance.

through the steel stud web causes many problems in steel-framed constructions.

We used modeling to compare three wall configurations (see Figure 3) to the traditional design with a full stud web. The first one (shape A) was also a traditional design, but it had 1½-inch x 4 inch holes punched in the studs, 24 inches OC. The next two shapes (B and C) represent the so-called expanded-channel design.

The stud web area was reduced by 11% in the shape A stud walls, by 63% in the shape B stud walls, and by 39% in the shape C stud walls. The

section area of the center of the stud web was reduced by 16% for shape A and by 87.5% for shapes B and C. A previous study made by National Research Council (NRC) Canada found a 50% reduction of the thermal bridge effect in walls with shapes similar to shapes B and C, compared with regular steel stud walls.

Clearly, walls with a reduced stud web area are much more thermally efficient than walls with traditional studs. The lowest framing factors, or *f*-values, were found for the walls containing shape B studs. However, shape C studs are significantly stronger, and their thermal performance is only slightly lower than that of shape B studs. With shape C studs, the stud web area was reduced about half as much as it was with shape B studs. Since walls containing studs B and C have similar thermal performances, building with shape C studs would be preferable, due to their superior structural stability. These data suggest important advantages to using punched studs; however, the lower structural integrity of punched-stud walls must also be taken into account, so more research is needed.

Novel Stud Design

Another way of minimizing steel stud web heat transfer is by replacing the steel web with a less conductive material, such as plywood or oriented strand board

(OSB). The FSEC wall cavity can be insulated with R-11 or R-13 fiberglass batts. For our hot-box tests, the exterior surface of the wall was finished with a ½-inch-thick layer of gypsum board to simulate an exterior insulated finish system (EIFS). The interior surface of the wall was

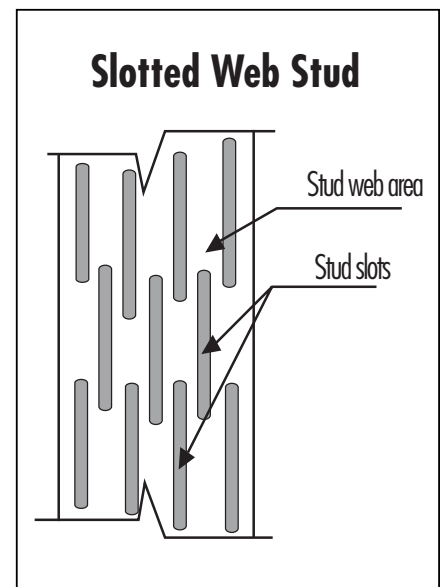


Figure 5. A series of slots in the stud web reduces the heat transfer area.

finished with a ½-inch-thick layer of gypsum board. Using the FSEC studs resulted in a 39% improvement in R-value, compared to using a traditional stud.

An interesting design of stud web has been proposed in Scandinavia. As shown in Figure 5, the web area is divided by several courses of slots. They significantly reduce effective heat conduction area on the stud web.

Stud Snugglers

Using insulation spacing between the studs and the sheathing is another way to reduce heat loss. Such insulation also reduces transverse heat transfer through the stud flanges. This kind of heat transfer increases heat loss in steel-framed structures. In 1993, our team at ORNL developed Studd Snugglers—foam shapes to cover

Table 1. R-Value and Framing Effect (*f*)* in 3½-Inch Steel Stud Walls

Exterior sheathing	R in Center of Cavity (in hft²F/Btu)	R test (in hft²F/Btu)	Framing effect (<i>f</i>)
½-in plywood	12.8	7.9	38.2%
½-in EPS	15.2	11.4	25.1%
1-in EPS	17.6	13.7	21.1%
2-in EPS	23.0	18.9	17.8%
2-in EPS, no cavity insulation	20.69	18.00	13.0%

* $f = 1 - (R_{\text{test}} / R_{\text{in center of cavity}}) \times 100\%$

Table 2. Thermal Performance of Walls with 1-in. Thick Stud Snugglers

Wall construction	Tested R-value hft ² F/Btu	Improvement hft ² F/Btu	Improvement %	Framing Effect %
Gypsum board, traditional 3 ⁵ / ₈ -in. studs, R-11 batts, gypsum board	7.9			38.2
Gypsum board, traditional 3 ⁵ / ₈ -in. studs, 1-in. foam profiles on studs, R-19 batts, gypsum board	16.3	8.4	106.3	13.0

the studs, (see Figure 6).

These shapes add highly efficient thermal insulation only in the locations where it is absolutely necessary (that is, in the stud flange areas). At the same time, the wall cavity is insulated with fiberglass batts, which are significantly less expensive than rigid foam sheathing. This reduces the thermal bridge effect at a relatively low cost. Similar technology was developed in Finland for steel trusses. This idea was later adopted in the United States in the form of so-called snap caps, which are foam caps that attach to the stud flanges with an adhesive.

ORNL built and tested steel stud walls containing the Stud Snugglers. In these walls, 1-inch-thick foam shapes covered the studs only in locations where strong thermal shorts were generated by the steel stud. We compared the R-value of this wall to that of walls made with conventional 3¹/₂-inch steel studs (see Table 2). Steel stud walls containing local stud insulation show excellent performance, both thermally and structurally. With its simplicity, high R-value

(R-16), low *f*-value (13%), and low cost, this system proves that, with proper thermal design, a steel stud wall can perform as well as a wood frame wall.

Steel Studs as Effective as Wood

Here at ORNL's BTC, we have developed two energy-efficient steel stud wall technologies in the last

two years. Our goal—to beat the performance of traditional 2 x 6 wood stud walls—was achieved in both cases. To make the walls economically attractive compared to traditional wood-framed construction; both new walls use only fiberglass insulation. No foam sheathing was necessary to reach or exceed R-18, the R-value of 2 x 6 wood stud walls.

The first wall uses 100% conventional materials. It can be built by any builder using the local Home Depot as a supply point. The R-value of this wall is about 19. The second wall uses a novel shape of steel stud (watch future issues of Home Energy for more information on this). The R-value of this wall is about 18. Both novel walls have the following advantages over conventional steel stud

walls:

- high R-value;
- very good acoustic insulation (the wall structure does not transmit vibrations);
- high fire resistance;
- easy assembly; and
- lack of foam sheathing.

Both designs were done in consultation with the North American Steel Framing Alliance. They are 100% doable, and the technologies can easily be adopted by any steel framing fabricator. For both walls, patents are pending.

Thermal Efficiency Achievable

There are various efficient means of reducing whole-building energy consumption in steel-framed buildings. Modifying the thermal envelope to optimize the material configuration and using the proper amount of thermal insulation will reduce first costs. Such savings can be easily achieved in the design stage, when materials are chosen. We found that adding rigid sheathing insulation to the wall is not always cost-effective. Some insulating techniques—such as dimples and thin foam strips installed on steel studs—generate only insignificant improvements of wall thermal performance. On the other hand, local insulation profiles located around steel studs or expanded-channel studs are effective ways of increasing the R-value of steel stud walls. The thermal efficiency of wood stud walls can be reached and exceeded with steel-framed technologies. Newly developed ORNL wall technologies are very good examples of this. 

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Stud Snuggler

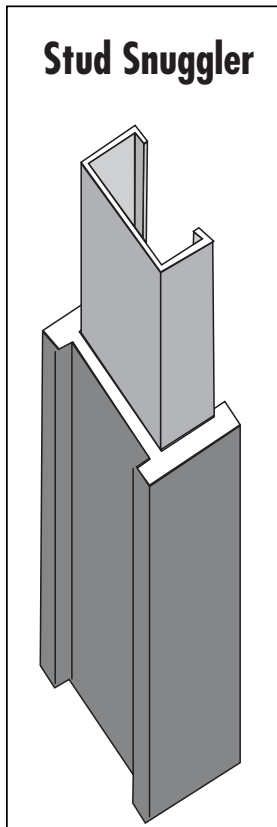


Figure 6. An interlocking foam cover wrapped around the steel stud protects the interior from intense heat transfer.

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